

Project Documentation

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AI Interview Question Generator (Interview Simulation)

Introduction

The AI Interview Question Generator is a desktop Python application built to simulate a real interview environment using Artificial Intelligence. The system analyzes resumes and job descriptions to generate interview questions based on hard skills, soft skills, experience level, job role, and selected difficulty level. It aims to provide a realistic interview practice experience for users preparing for technical and professional job interviews.

Problem Statement

Many students and job seekers struggle to prepare effectively for interviews due to a lack of personalized practice and realistic interview experience. Existing preparation methods often provide generic questions that do not match individual skills, experience, or job roles. This project aims to solve this problem by developing an AI-powered interview system that generates personalized interview questions based on resumes and job requirements.

Objectives

- Develop an AI-powered interview simulation platform.
- Automatically analyze resumes and extract skills and experience.
- Identify and separate hard skills and soft skills.
- Generate interview questions dynamically using AI.
- Support multiple difficulty levels (Beginner, Intermediate, Expert).
- Enable voice interaction through Speech-to-Text (STT) and Text-to-Speech (TTS).

System Features

1. Resume Analysis

The system extracts information such as:

- Hard skills (Python, MySQL, etc.)
- Soft skills (communication, teamwork, etc.)
- Experience level
- Projects and technologies used

2. Job Description Analysis

The application analyzes job requirements to identify:

- Required hard skills
- Required soft skills
- Experience expectations

3. AI Question Generation

Questions are generated dynamically based on:

- Hard skills (Python, MySQL, etc.)
- Soft skills (communication, teamwork, etc.)
- Experience level

4. Interview Simulation Mode

The AI acts as an interviewer by:

- Asking questions one at a time
- Accepting responses through text or voice
- Dynamically moving to the next question
- Saving responses for evaluation and review

5. Difficulty Levels

- **Beginner:** Basic concepts, definitions, simple syntax/problems, and foundational understanding
- **Intermediate:** Practical problem-solving, real-world scenarios, optimization, and applying concepts together.
- **Expert:** Deep technical trade-offs, architecture, scalability, edge cases, and advanced debugging/strategy.

6. Voice Interaction

- Speech-to-Text for answering through voice
- Text-to-Speech for reading interview questions aloud

Technologies Used

- **Python:** Core programming language
- **Tkinter:** GUI framework for desktop application
- **spaCy:** Natural Language Processing
- **Gemini API:** AI-powered question generation
- **MySQL:** Database management
- **PyPDF2:** Resume parsing
- **SpeechRecognition:** Speech-to-Text
- **gTTS:** Text-to-Speech

System Workflow

1. User uploads a resume.
2. The system extracts skills and experience.

3. The user selects a job role and difficulty level.
4. Job requirements are analyzed.
5. AI generates interview questions based on skills and level.
6. Interview simulation begins.
7. User answers through text or voice.
8. Responses are stored in the database for review.

Work Division

1. Member 1 (Prasanna Shakya): UI Designer & Frontend Engineer

Primary Tools: Tkinter

Responsibilities:

- Design and build the desktop windows (Login, Resume Upload, Interview Room, Dashboard).
- Manage screen transitions and UI states (loading spinners, text boxes, buttons).
- Display the real-time conversation transcript and final scorecard on the screen.

2. Member 2 (Navaraj Thapa): Audio & Media Engineer

Primary Tools: SpeechRecognition, gTTS

Responsibilities:

- Configure microphone inputs and manage audio hardware connection using SpeechRecognition.
- Convert the user's spoken answers into text strings to pass to the Backend.
- Take the AI text responses and convert them to spoken audio using gTTS.
- Handle background thread management so the UI doesn't freeze while audio is playing/recording.

3. Member 3 (Saksham Smith Sunar): Backend & Integration Engineer

Primary Tools: Python Core, System Architecture

Responsibilities:

- Act as the "glue" connecting Member 1 (UI), Member 2 (Audio), and Member 4 (Database).
- Manage the interview flow state (e.g., tracking current question count, session IDs).
- Pass data between the front-end inputs and the back-end AI processing scripts.

4. Member 4 (Samyam Dhaubaji Shrestha): Database Administrator (DBA) & Security

Primary Tools: MySQL

Responsibilities:

- Design the relational database schema (User, Interviews, Questions, Transcripts, Scores tables).
- Write optimized SQL queries for saving transcripts and fetching historical interview performance.
- Securely manage API keys (Gemini) and user authentication data locally.

5. Member 5 (Eroj Bajracharya): AI, NLP & Data Specialist

Primary Tools: Gemini API, spaCy, PyPDF2

Responsibilities:

- Build the resume parser module using PyPDF2 and clean text with spaCy.
- Extract candidate keywords and skills from the resume using spaCy's entity extraction.

- Engineer the Gemini system prompts to generate questions based on the resume and grade the final transcript.

Development Plan

- **Week 5:** Member 5 builds the spaCy/PyPDF2 parser and tests it. Member 4 provisions the MySQL database on a local server or free cloud tier.
- **Week 6:** Members 1 and 2 build the Tkinter GUI and configure the SpeechRecognition/gTTS audio loops independently. Member 3 simultaneously connects the Gemini API to local Python scripts using the system prompt above.
- **Week 7:** Finalize the project by connecting everything, and doing testing.
- **Week 8:** Spend the last days doing system testing and polishing the presentation.

Learning Expectations

Through the development of the AI Interview Question Generator, we expect to gain practical experience in software development, artificial intelligence, and system integration. As a team, we aim to strengthen our programming skills in Python while learning how to apply technologies such as Natural Language Processing (NLP), database management, speech recognition, and AI-powered question generation in a real-world application. We also expect to improve our teamwork, communication, problem-solving, and project management abilities by collaborating across different areas of development. Additionally, this project will help us understand the complete software development lifecycle, including planning, designing, testing, debugging, and integrating multiple technologies to build an intelligent system that addresses a real-world problem.

Conclusion

The AI Interview Question Generator is an intelligent interview preparation system that combines AI, resume analysis, and voice interaction to deliver a realistic interview experience. By generating personalized questions based on skills, experience, and job requirements, the platform helps users improve interview performance and strengthens technical and professional readiness.